

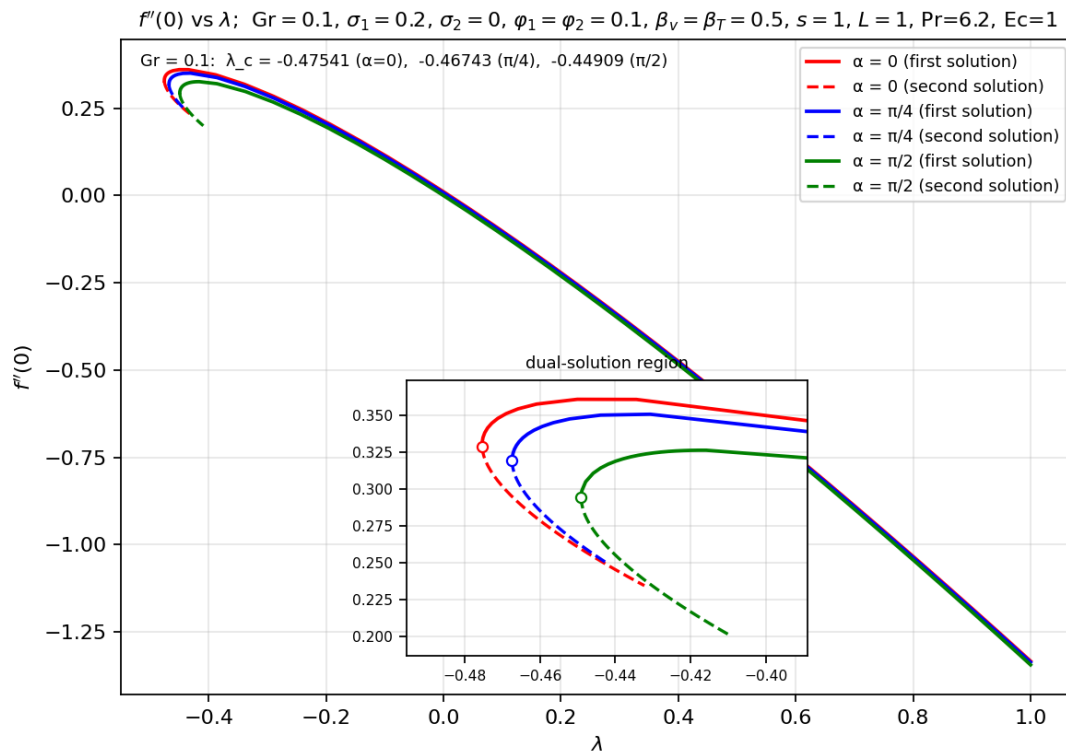
Dhia Problem 3 — Results v6 (PROMPT 4): all results regenerated with $\sigma_2 = 0$

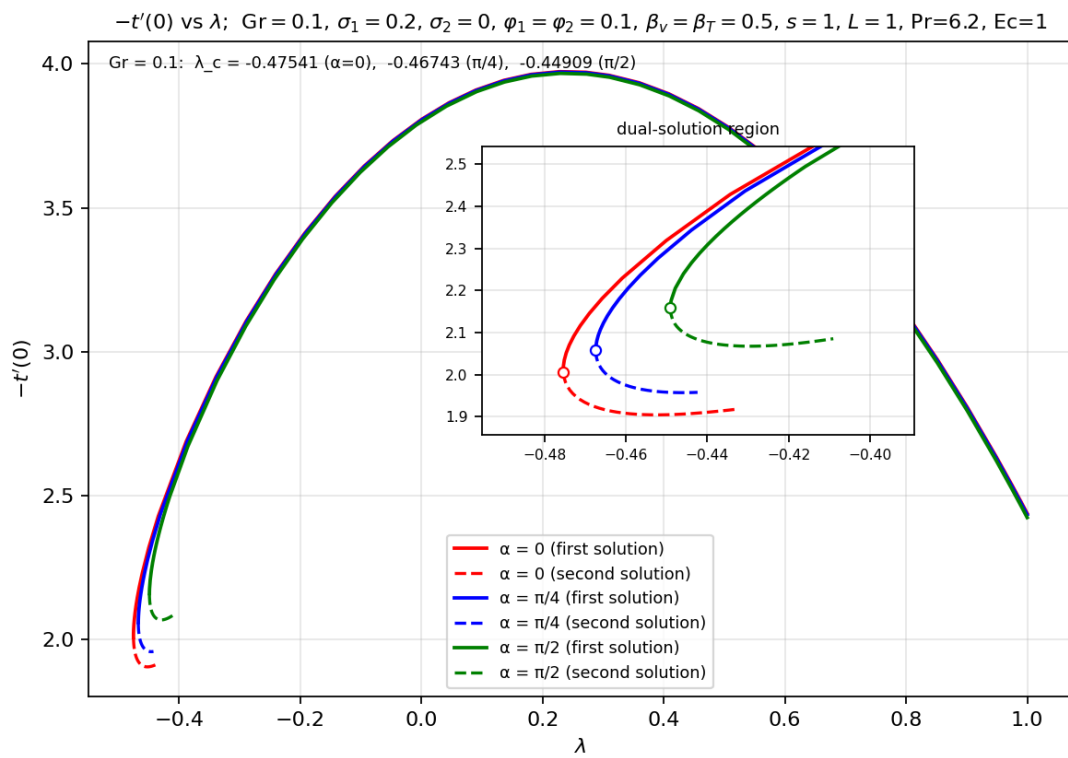
Parameters: $\sigma_2 = 0$, $\phi_1 = \phi_2 = 0.1$, $\beta_v = \beta_T = 0.5$, $s = 1$, $Pr = 6.2$, $Ec = 1$, $m = 1$, $\varepsilon = 1$, $L = 1$ ($\sigma_1 = 0.2$, the program default, unchanged). The inclination angle is written as α in every figure. As established previously, the program's $Gr = 0.001$ makes the three α curves coincide, so the two agreed Gr sets are used: **$Gr = 0.1$** (primary; complete dual structures for all three α) and **$Gr = -0.5$** (opposing flow, strongest separation). $\alpha = \pi/2$ is identical for every Gr since $\cos(\pi/2) = 0$.

Set	Gr	lambda_c ($\alpha=0$)	lambda_c ($\alpha=\pi/4$)	lambda_c ($\alpha=\pi/2$)
Primary	0.1	-0.47541	-0.46743	-0.44909
Opposing	-0.5	-0.34228	-0.37058	-0.44909

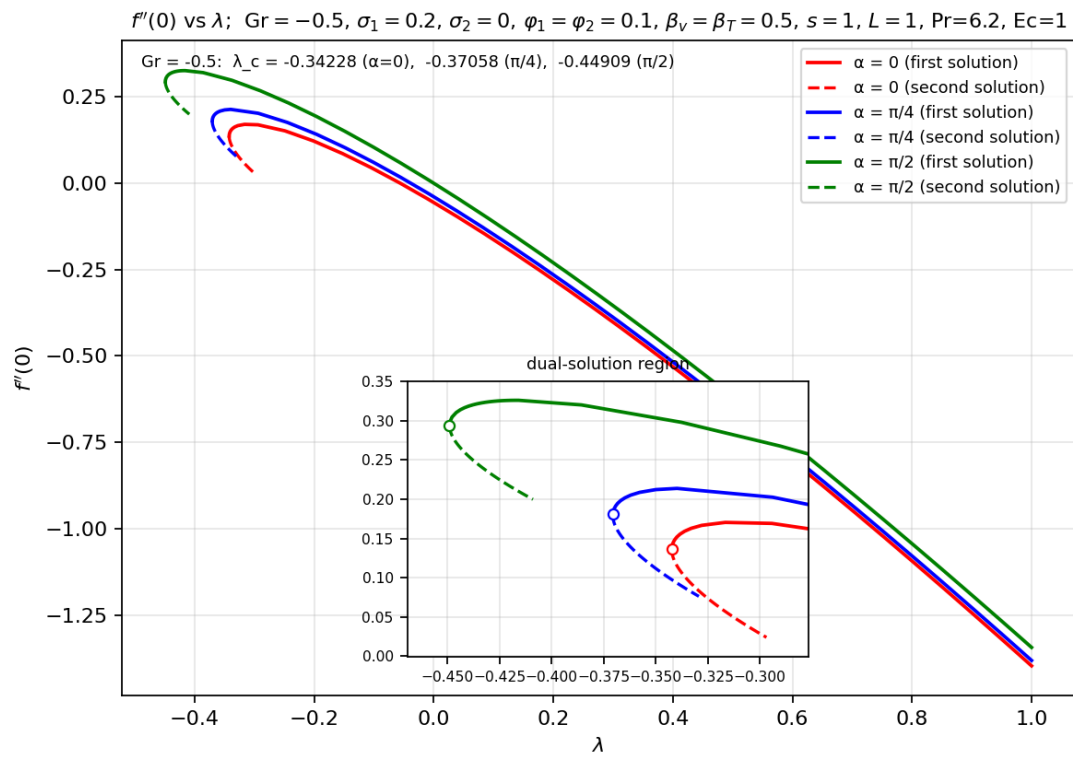
Compared with the $\sigma_2 = 0.2$ results, the critical values shift slightly (e.g. $Gr = 0.1$, $\alpha=0$: -0.47109 $\rightarrow$ -0.47541) and the $-t'(0)$ level rises because the wall temperature is now fixed at $t(0) = 1$ (no thermal slip). Full curve data: task2_curve_*.csv.

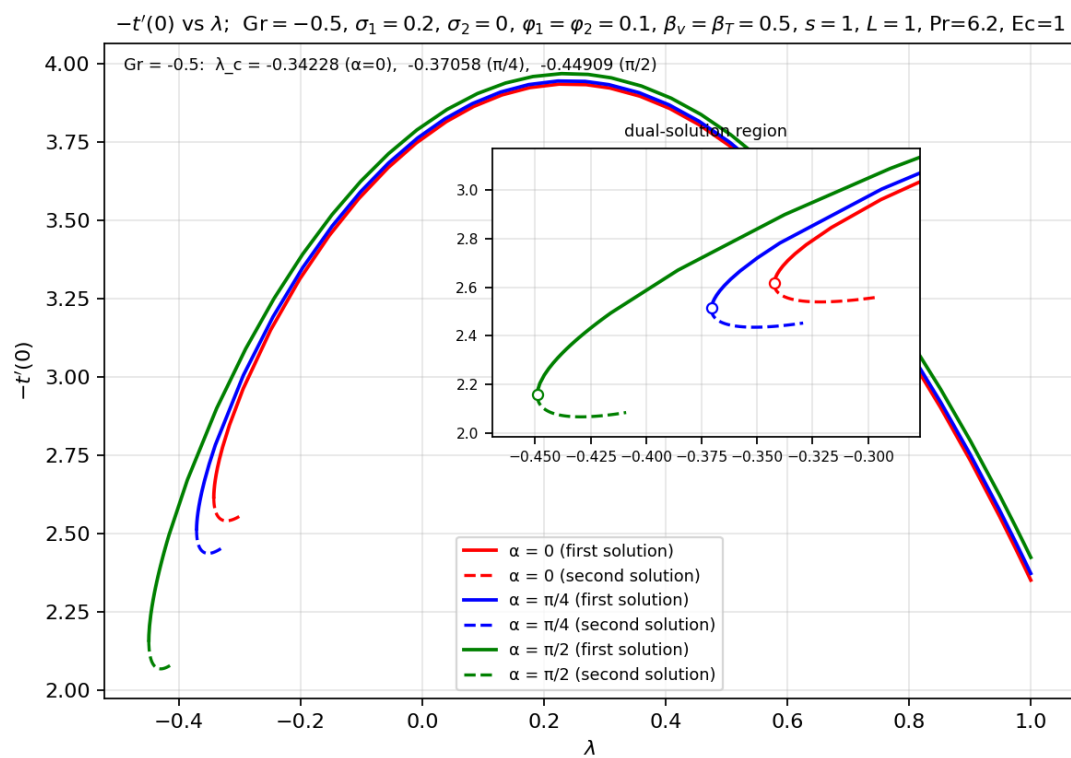
$f''(0)$ and $-t'(0)$ versus λ — $Gr = 0.1$





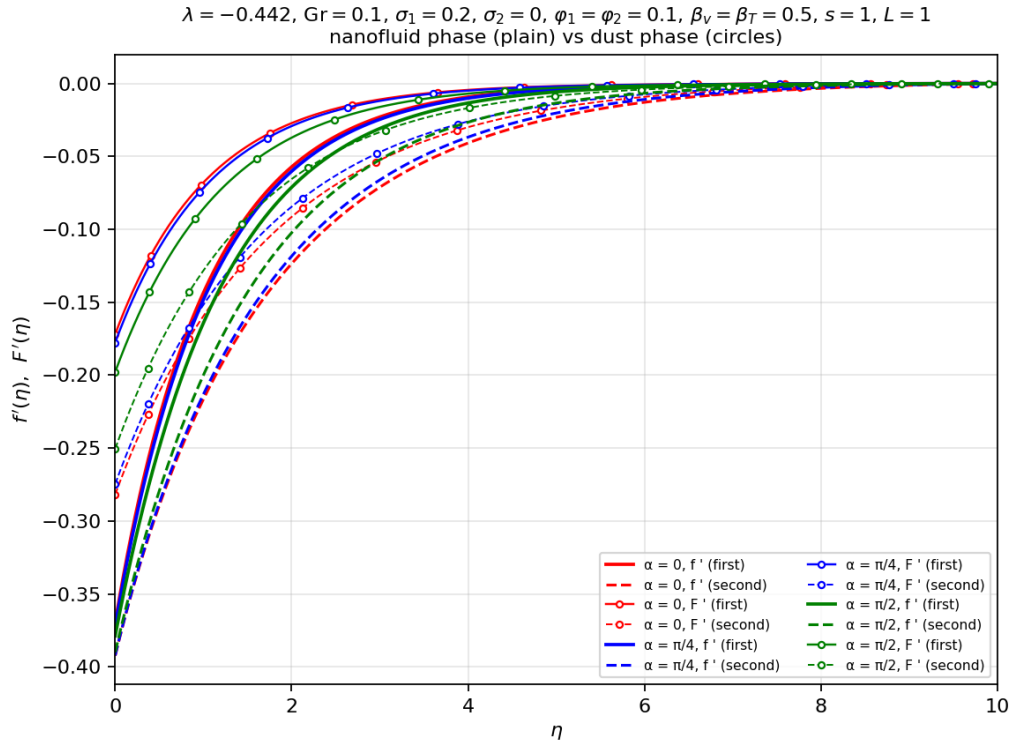
$f''(0)$ and $-t'(0)$ versus λ — $Gr = -0.5$ (opposing flow)



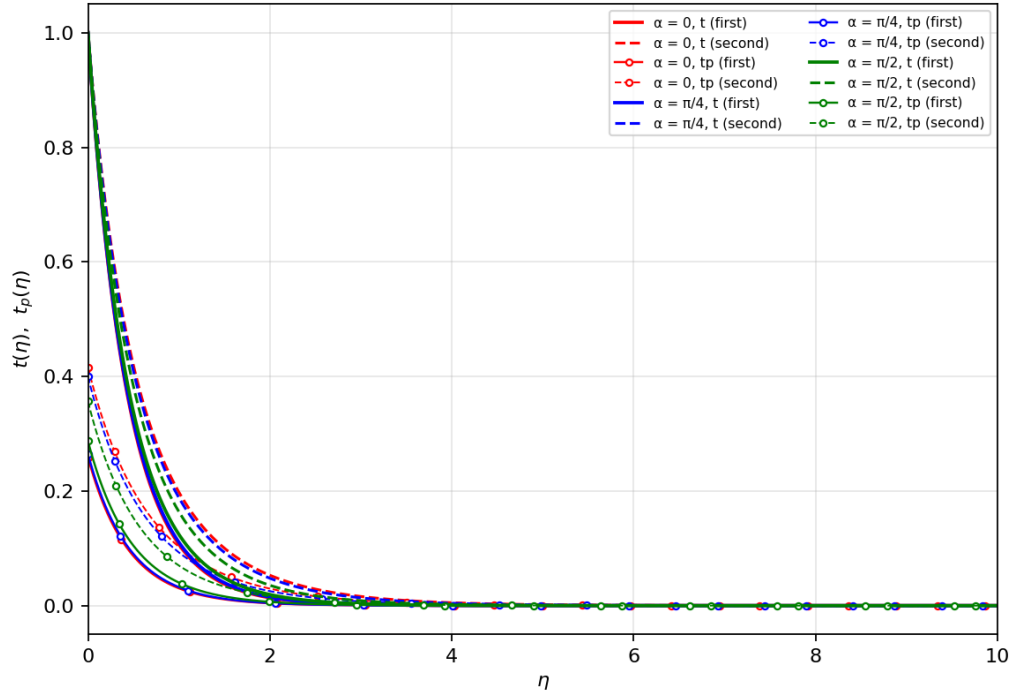


Dual velocity and temperature profiles ($\lambda = -0.442$, $Gr = 0.1$, $\sigma_2 = 0$)

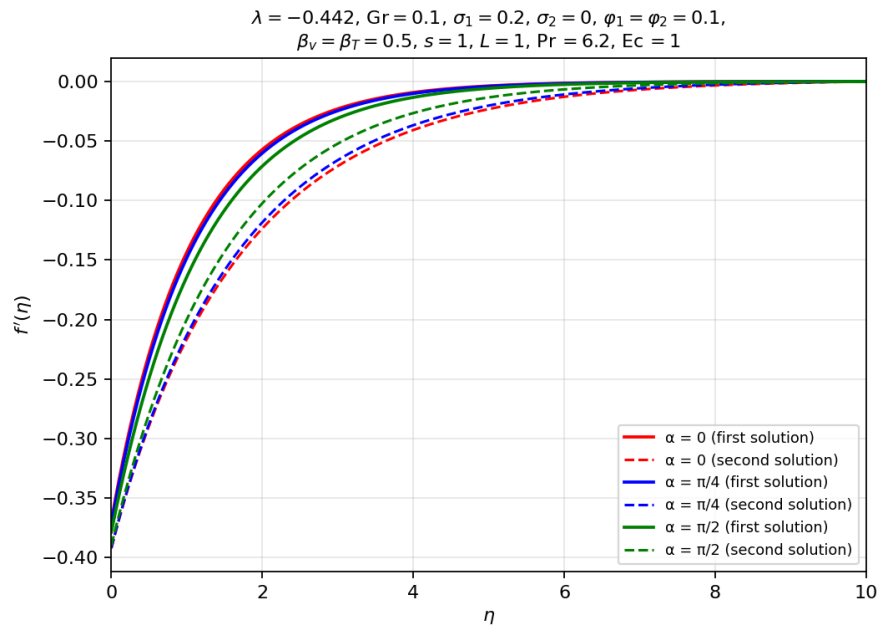
$\lambda = -0.442$ lies inside the common dual-solution window of all three inclinations ($\alpha=0$: $\lambda_c = -0.4754$; $\pi/4$: -0.4674 ; $\pi/2$: -0.4491). Solid = first solution, dashed = second solution; every x-axis starts at $\eta = 0$. All OdeBC boundary conditions are satisfied ($f(0)=s$, $f'(0)=\lambda+\sigma_1 f''(0)$, $t(0)=1$ since $\sigma_2=0$, and all profiles decay to zero far from the wall).



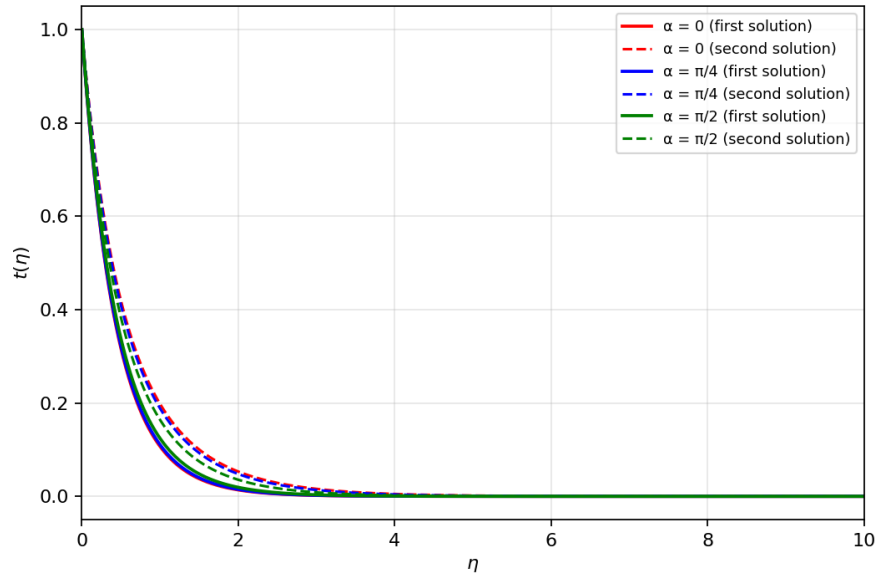
$\lambda = -0.442$, $Gr = 0.1$, $\sigma_1 = 0.2$, $\sigma_2 = 0$, $\varphi_1 = \varphi_2 = 0.1$, $\beta_V = \beta_T = 0.5$, $s = 1$, $L = 1$
nanofluid phase (plain) vs dust phase (circles)

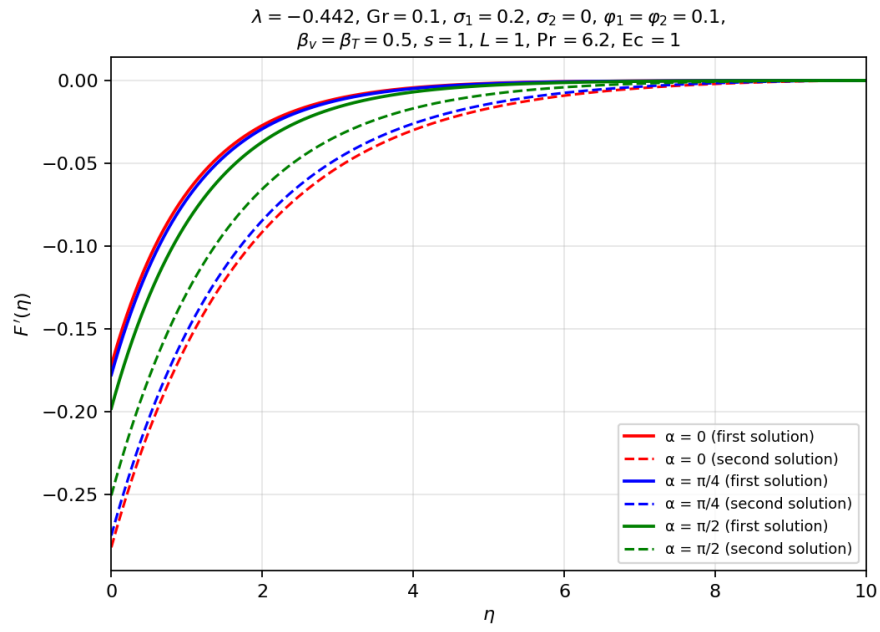


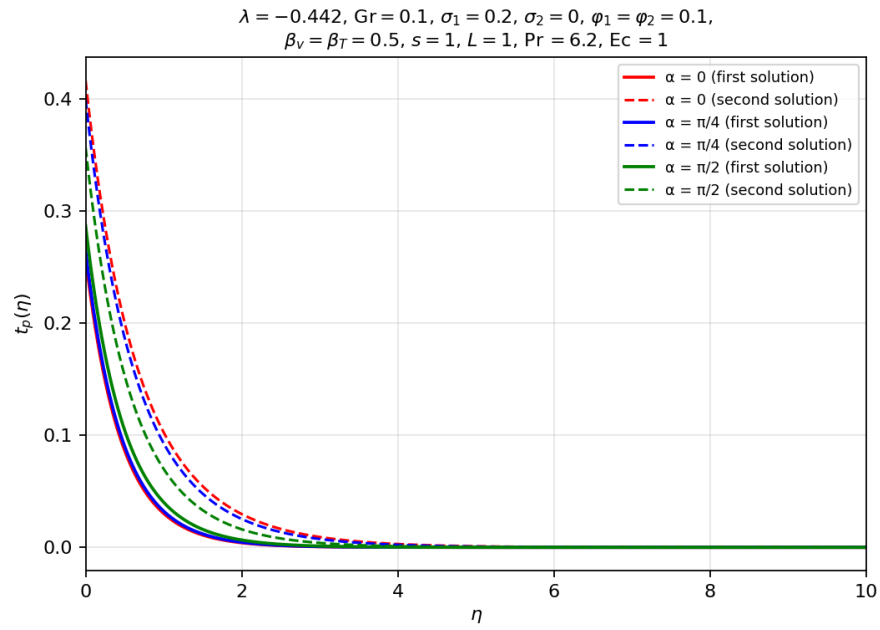
Individual profiles



$\lambda = -0.442$, $Gr = 0.1$, $\sigma_1 = 0.2$, $\sigma_2 = 0$, $\varphi_1 = \varphi_2 = 0.1$,
 $\beta_V = \beta_T = 0.5$, $s = 1$, $L = 1$, $Pr = 6.2$, $Ec = 1$







α	solution	$f''(0)$	$F'(0)$	$-t'(0)$	$tp(0)$
0	first	0.361010	-0.171929	2.377075	0.258637
0	second	0.248039	-0.281814	1.908498	0.416177
$\pi/4$	first	0.350122	-0.177633	2.357621	0.264635
$\pi/4$	second	0.249128	-0.274683	1.957632	0.399618
$\pi/2$	first	0.316444	-0.197831	2.287122	0.286971
$\pi/2$	second	0.260150	-0.250494	2.082085	0.356457

Full profile data: task3_profile_first/second_X*.csv in this folder.